

COURSE NAME: COMPUTER VISION (ITA05)

COURSE OUTCOME: CO4 - Object & Pattern Recognition Techniques (BTL 5)

WEIGHTAGE: 40%

EVALUATION: Detailed Technical Justifications

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints: Viola-Jones (Fast, 78% acc), YOLO (Moderate speed, 92% acc), Deep Learning (Slow, 95% acc). Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy.

COMPARATIVE ANALYSIS & EVALUATION

- **Viola-Jones:** Uses Haar-like features and AdaBoost cascades. Extremely low computational overhead, but low accuracy (78%) and prone to false positives under non-standard lighting or occlusion.
- **Deep Learning (Heavy CNNs):** Achieves state-of-the-art accuracy (95%), but suffers from high inference latency and computational overhead, making it unsuitable for edge real-time deployment without specialized hardware.
- **YOLO (You Only Look Once):** Frames object detection as a single regression problem, striking an optimal operational balance with 92% accuracy and sub-50ms latency.

FINAL DECISION & JUSTIFICATION

Selected Model: YOLO. Under strict real-time constraints (e.g., target <50ms response time), Viola-Jones fails safety-critical accuracy thresholds (<80%), whereas heavy Deep Learning architectures violate frame-rate requirements (causing frame drops). YOLO provides high detection precision (92%) while maintaining high frame rates, making it the most robust candidate for real-world surveillance deployment.

2. Robust Detection under Environmental Variations

Detection accuracy: Viola-Jones (Normal: 80%, Occlusion: 60%, Low Light: 55%), YOLO (Normal: 90%, Occlusion: 75%, Low Light: 70%), Deep Learning (Normal: 95%, Occlusion: 85%, Low Light: 80%). Compare robustness and evaluate best overall performance across varying conditions.

Method	Normal	Occlusion	Low Light	Max Accuracy Drop
Viola-Jones	80%	60%	55%	25%
YOLO	90%	75%	70%	20%
Deep Learning	95%	85%	80%	15%

ROBUSTNESS ANALYSIS

Viola-Jones exhibits severe degradation under environmental stress (dropping from 80% to 55%), as handcrafted pixel-intensity features are highly vulnerable to shadows and partial coverage. YOLO maintains acceptable performance but drops by 20% in low light due to grid-cell feature dispersion. Deep Learning (deep CNNs) exhibits the highest stability, with only a 15% performance drop under non-ideal environments.

FINAL DECISION & JUSTIFICATION

Selected Model: Deep Learning (Deep CNNs). For applications where environmental variability (occlusions, changing lighting) is critical, Deep Learning provides superior feature abstraction and multi-layer invariance, sustaining 80%+ accuracy across all stress conditions.

3. Optical Flow Reliability vs Noise Sensitivity

Method A: Stable vectors but misses small motions. Method B: Detects fine motion but noisy vectors. Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes.

COMPARATIVE ANALYSIS

- **Method A (Dense / Coarse Regularized Flow e.g., Horn-Schunck with heavy smooth constraint):** High spatial smoothness filter suppresses noise, ensuring global tracking stability, but smooths out localized gradient changes required to capture subtle movements.
- **Method B (Sparse / High Sensitivity Flow e.g., Lucas-Kanade with small window size):** Highly sensitive to subtle optical intensity variations, but highly susceptible to sensor noise, aperture problems, and background movement.

FINAL DECISION & JUSTIFICATION

Selected Model: Method A (or Method B integrated with Kalman / Particle Filtering). In general dynamic scene tracking, spurious motion vectors (false motion signals) destabilize trajectories significantly more than minor temporal missing events. Method A is preferred for macro-object trajectory estimation due to tracking stability. If fine micro-motion tracking is strictly required, Method B must be paired with temporal noise filtering.

4. Background Modeling Strategy (GMM)

A GMM-based system shows high false positives in dynamic background and missed detections in static background. Compare performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

SCENARIO ANALYSIS

- **Dynamic Background (Swaying trees, water ripples):** High false positives occur because standard K-Gaussian distributions fail to quickly adapt variance parameters to multi-modal rapid intensity fluctuations.
- **Static Background (Camouflaged or slow-moving objects):** Missed detections occur when a slowly moving target gets absorbed into the background Gaussian model due to static learning rates.

EVALUATION & PROPOSED IMPROVEMENTS

GMM alone is insufficient. To resolve these limitations:

1. **Adaptive Learning Rate (α):** Implement dynamic learning rates that slow down update frequency for stationary foreground pixels to prevent background absorption.
2. **Hybrid Spatial-Temporal Filtering:** Integrate Spatial Texture Descriptors (e.g., Local Binary Patterns - LBP) or Deep Optical Flow to distinguish dynamic background noise from true semantic foreground.

5. Motion Estimation vs Computational Cost

Method A: Low error (5%), high computation. Method B: Moderate error (10%), low computation. Compare methods and evaluate suitability for real-time vs offline applications.

COMPARATIVE ANALYSIS

Method A prioritizes spatial-temporal precision at the cost of execution time (e.g., iterative non-linear optimization or 3D CNNs). Method B optimizes for throughput and operational efficiency (e.g., block-matching motion estimation or fast pyramidal Lucas-Kanade) at a minor accuracy trade-off (5% error margin difference).

APPLICATION MAPPING & JUSTIFICATION

- **Real-Time Applications (Autonomous navigation, video conferencing):** Select **Method B**. The 10% error margin is acceptable given the strict requirement for frame-rate consistency without buffer accumulation or latency degradation.
- **Offline Applications (Forensic analysis, medical imaging motion correction):** Select **Method A**. Computation time is not a constraining factor, making maximum tracking accuracy (95% precision) the paramount design criteria.

6. Structure from Motion (SfM) Consistency

Dataset A: Accurate depth but requires high-quality images. Dataset B: Inconsistent depth but works on low-quality images. Compare datasets and evaluate which setup is more practical for real-world deployment.

COMPARATIVE ANALYSIS

Dataset A relies on dense, high-resolution feature correspondences (e.g., crisp edge descriptors), rendering high metric fidelity but failing outright in low-resolution, noisy real-world imagery. Dataset B operates reliably across sub-optimal capture devices (e.g., low-cost mobile sensors) but exhibits metric depth jitter and reconstruction noise.

PRACTICAL DEPLOYMENT EVALUATION

Selected Setup: Dataset B paired with Post-Processing Optimization (or Dataset A depending on operational control). In uncontrolled field conditions (surveillance, consumer mobile apps), capturing high-quality images consistently is infeasible, making Dataset B pipelines more robust to operational failure. However, for industrial inspection, controlled capture setups using Dataset A are preferred. For general deployment, high system availability using Dataset B with temporal filtering yields the highest utility.

7. Stereo Vision vs Motion Estimation

Stereo Vision: Accurate but requires dual cameras. Motion Estimation (Monocular Depth / SfM): Less accurate but works with a single camera. Compare both and evaluate under hardware constraints.

Feature	Stereo Vision	Motion Estimation (Monocular)
Hardware Requirement	Dual synchronized cameras, fixed baseline	Single standard camera
Accuracy & Scale	High absolute metric depth accuracy	Relative depth, scale ambiguity
Computational Load	Epipolar matching (Moderate)	Temporal feature tracking / Ego-motion (High)

FINAL DECISION & JUSTIFICATION

Under Strict Hardware Constraints (Cost/Size/Power): Select Motion Estimation. In miniaturized or budget-constrained systems (e.g., low-cost drones, single-camera IoT nodes), hardware elimination takes priority. **Under Safety-Critical Real-Time Constraints: Select Stereo Vision.** Stereo vision provides immediate instance frame depth via geometric triangulation without requiring ego-motion or temporal delay.

8. Model Generalization vs Overfitting

Model A: Training 98%, Testing 70%. Model B: Training 90%, Testing 88%. Compare models and evaluate which is more reliable for deployment.

COMPARATIVE ANALYSIS

- **Model A:** Exhibits severe variance / overfitting (28% performance gap between train and test). It has memorized training noise and will fail on unseen real-world distributions.
- **Model B:** Exhibits high generalization stability (2% gap between train and test) with an overall high test performance of 88%.

FINAL DECISION & JUSTIFICATION

Selected Model: Model B. Model B is significantly more reliable for field deployment. Model A poses severe operational risk due to non-generalizable feature learning. Model B's minimal generalization gap guarantees consistent performance when presented with real-world out-of-distribution sample data.

9. YOLO Performance across Scales

Detection results: Large objects (95%), Medium objects (85%), Small objects (60%). Compare performance across scales, evaluate if YOLO alone is sufficient, and propose improvements.

SCALE PERFORMANCE ANALYSIS

Standard single-stage detectors like early YOLO struggle with small objects (60% accuracy) due to downsampling operations in deep convolutional layers where fine spatial details are lost in coarse grid cells.

EVALUATION & ARCHITECTURAL IMPROVEMENTS

YOLO alone (vanilla) is insufficient for multi-scale tasks. To achieve multi-scale competence, the following structural enhancements must be incorporated:

1. **Feature Pyramid Networks (FPN) / PANet:** Construct multi-scale feature hierarchies to concatenate high-level semantic features with low-level spatial features.
2. **Multi-Scale Testing & Higher Input Resolution:** Process images at higher spatial grids and utilize anchor-box optimization tuned for tiny aspect ratios.

10. Motion Model Selection for Animation

Model A: Smooth but physically inaccurate. Model B: Realistic but computationally intensive. Compare models and evaluate usage under real-time vs cinematic constraints.

COMPARATIVE ANALYSIS

Model A uses kinematic interpolation (e.g., splines, blendshapes) which guarantees visual smoothness and minimal computational latency but ignores true physical dynamics (gravity, inertia, multi-body collisions). Model B uses full physics-based numerical solvers (rigid/soft body dynamics, finite element analysis) delivering exact realism at high compute overhead.

APPLICATION MAPPING & JUSTIFICATION

- **Real-Time Animation (Interactive Gaming / VR): Select Model A.** Maintaining fixed target frame rates (60–120 FPS) without rendering stutter is paramount; visual plausibility takes precedence over exact physics.
- **Cinematic VFX & Offline Rendering: Select Model B.** Visual fidelity and exact physical realism are non-negotiable, and rendering can be distributed across offline render farms.

11. Detection Model Comparison under Resource Constraints

Performance: Model A (85% acc, 40ms), Model B (92% acc, 90ms), Model C (90% acc, 60ms). Compare models and evaluate optimal choice for real-time systems with moderate accuracy requirements.

Model	Accuracy	Latency	Throughput (FPS)	Efficiency Ratio (Acc/Latency)
Model A	85%	40 ms	25 FPS	2.125
Model B	92%	90 ms	11.1 FPS	1.022
Model C	90%	60 ms	16.6 FPS	1.500

FINAL DECISION & JUSTIFICATION

Selected Model: Model C. Model C provides the optimal sweet spot. It achieves a high accuracy of 90% (a 5% increase over Model A) while maintaining a processing time of 60ms (~16.6 FPS, satisfying real-time constraints). Model B violates real-time limits (90ms leads to frame lag under typical camera processing pipelines), whereas Model A needlessly sacrifices 5% accuracy for unrequired latency overhead reduction.

12. Robustness vs Speed Trade-off

Performance: Method A (88% acc, High Rob, Slow), Method B (85% acc, Mod Rob, Fast), Method C (90% acc, Low Rob, Moderate). Compare methods and evaluate best choice for a balanced system.

COMPARATIVE TRADE-OFF ANALYSIS

Method C achieves the highest peak laboratory accuracy (90%) but fails in real-world deployment due to low environmental robustness. Method A offers exceptional robustness but suffers from computational latency (Slow). Method B maintains high speed (Fast) with acceptable accuracy (85%) and moderate robustness.

FINAL DECISION & JUSTIFICATION

Selected Model: Method B. For a general balanced system deployed in active environments, low robustness (Method C) is an immediate single-point-of-failure risk. Method B delivers reliable baseline performance across environments with the highest execution throughput, fulfilling balance objectives.

13. Optical Flow Stability Analysis

Method A: High accuracy, High noise sensitivity. Method B: Moderate accuracy, Low noise sensitivity. Compare both methods and evaluate suitability for real-time tracking in noisy environments.

COMPARATIVE ANALYSIS & EVALUATION

In noisy dynamic environments (e.g., outdoor rain, low-grade camera sensors), high noise sensitivity (Method A) causes random velocity vector jitter, leading to severe tracking drift and trajectory accumulation errors, despite its raw accuracy in ideal conditions.

FINAL DECISION & JUSTIFICATION

Selected Model: Method B. Stability in optical flow tracking under noisy operational conditions is governed by noise immunity. Method B's low noise sensitivity guarantees consistent spatial tracking vector direction over sequential frames, preventing catastrophic tracker drift.

14. Motion Estimation Accuracy vs Efficiency

Method A (Error: 4%, High Comp), Method B (Error: 9%, Low Comp), Method C (Error: 6%, Mod Comp). Compare methods and evaluate the best approach for embedded systems.

Method	Error Rate	Computational Cost	Embedded Viability
Method A	4%	High	Unsuitable (Power/Thermal Limits)
Method B	9%	Low	Highly Viable (High speed, minimal drain)
Method C	6%	Moderate	Optimal Balanced Choice

FINAL DECISION & JUSTIFICATION

Selected Model: Method C (or Method B under strict low-power constraint). Method C reduces the motion estimation error rate significantly to 6% (compared to B's 9%) while keeping resource overhead manageable. However, on micro-embedded platforms with strict thermal and energy budgets, Method B is preferred if 9% error is within functional tolerances. Overall, Method C offers the most effective accuracy-to-compute balance.

15. GMM Performance in Different Conditions

Results: Static Scenario (92% acc), Dynamic Scenario (68% acc). Compare performance and evaluate whether GMM is reliable for all environments. Suggest improvements.

PERFORMANCE COMPARISON

GMM performs exceptionally well in static environments (92% accuracy) because pixel distributions are unimodal or stationary multi-modal. In dynamic environments, rapid illumination or background variations (waving foliage, water reflections) degrade accuracy to 68% due to model saturation and static distribution parameters.

EVALUATION & PROPOSED ENHANCEMENTS

GMM is NOT reliable for all environments.

- **Shadow Removal Modules:** Integrate color-space conversion (HSV/Chrominance analysis) to separate intensity drops (shadows) from surface structural changes.
- **Spatiotemporal Texture Modeling:** Combine GMM with Local Binary Pattern (LBP) operators to verify structural texture consistency alongside color likelihood.

16. Object Detection under Scale Variation

Large objects: 94%, Medium objects: 80%, Small objects: 55%. Compare detection performance, evaluate system limitations, and justify improvements.

LIMITATION ANALYSIS

The systematic drop from 94% down to 55% highlights feature map resolution loss. Downsampling strides in standard CNN backbones cause spatial details of small objects to merge into single receptive fields, leading to missing detections.

JUSTIFIED SYSTEM IMPROVEMENTS

1. **Feature Pyramid Networks (FPN):** Build top-down feature aggregation paths to retain low-level high-resolution feature maps for small target detection.
2. **Tiling / Image Cropping Pipelines:** Slice high-resolution input frames into overlapping tiles during inference to increase relative pixel size of small instances.

17. Stereo vs SfM Accuracy Comparison

Stereo (High acc, Dual camera), SfM (Moderate acc, Single camera). Compare and evaluate which is more practical under hardware limitations.

COMPARATIVE ANALYSIS

Stereo requires a calibrated physical baseline and dual synchronized hardware sensors, increasing hardware footprint, power usage, and monetary cost. Structure from Motion (SfM) estimates 3D structure temporally from sequential frames captured using a single camera.

PRACTICALITY EVALUATION

Selected System: Structure from Motion (SfM). Under hardware restrictions (weight, footprint, cost, port availability), SfM eliminates the need for secondary camera hardware. Although metric accuracy is moderate, it achieves operational feasibility where dual-camera setups are physically or financially prohibitive.

18. Detection Model Generalization

Model A (Training: 97%, Testing: 75%), Model B (Training: 90%, Testing: 88%). Compare and evaluate which model is better for deployment.

COMPARATIVE EVALUATION

Model A suffers from high variance (overfitting gap of 22%), demonstrating poor generalization to real-world out-of-sample data. Model B displays excellent variance control (overfitting gap of only 2%) and achieves a vastly superior test accuracy (88% vs 75%).

FINAL RECOMMENDATION

Selected Model: Model B. Model B guarantees stable performance in actual operational deployment, drastically reducing unexpected detection failures on real-world test distributions.

19. Motion Tracking Accuracy Comparison

Method A (Accuracy: High, Stability: Low), Method B (Accuracy: Moderate, Stability: High). Compare and evaluate which is suitable for long-term tracking.

COMPARATIVE ANALYSIS

Long-term tracking systems accumulate spatial uncertainty over consecutive frames ($t_1, t_2, \dots, t_n$). High accuracy paired with low stability (Method A) leads to periodic tracking failures, bounding-box jitter, and tracklet re-identification loss.

FINAL DECISION & JUSTIFICATION

Selected Model: Method B. High trajectory stability is the governing requirement for long-term tracking to maintain object identity continuity and avoid accumulated drift errors over extended frame sequences.

20. Surveillance Model Selection

Model A (82% acc, 30ms latency), Model B (90% acc, 70ms latency), Model C (95% acc, 120ms latency). Compare and evaluate under strict latency constraints.

LATENCY CONSTRAINT EVALUATION

Assuming standard real-time video streaming constraints (e.g., target < 50ms latency / 20+ FPS):

- **Model C (120ms):** Processes at ~8.3 FPS, violating real-time playback buffers and causing significant delay.
- **Model B (70ms):** Processes at ~14.2 FPS, exceeding the 50ms strict threshold.
- **Model A (30ms):** Processes comfortably at ~33.3 FPS.

FINAL DECISION & JUSTIFICATION

Selected Model: Model A. Under strict real-time surveillance latency constraints (<50ms), Model A is the only viable candidate that prevents video buffer lag while maintaining acceptable precision (82%).

21. Noise Impact on Detection

Noise Level vs Accuracy: Low (92%), Medium (80%), High (60%). Compare performance and evaluate system robustness.

ROBUSTNESS ANALYSIS

The system displays high sensitivity to input sensor noise, with detection accuracy degrading by 32% under severe noise. High-frequency pixel noise corrupts gradient descriptors and edge features essential for object feature extraction.

SYSTEM MITIGATION STRATEGIES

To improve system robustness in real-world deployments:

1. **Preprocessing Denoising Filters:** Embed fast Gaussian or Bilateral filtering directly in the camera ingestion pipeline.
2. **Noise Augmentation:** Retrain models using Gaussian and Salt-and-Pepper noise data augmentation to build feature map invariance.

22. Depth Estimation Error Analysis

Method A (0.5m error), Method B (1.2m error), Method C (0.8m error). Compare and evaluate accuracy for real-world applications.

Method	Mean Depth Error	Precision Rating	Application Suitability
Method A	0.5 m	High	Autonomous Driving, Robotics, Docking
Method C	0.8 m	Moderate	General Spatial Mapping, Augmented Reality
Method B	1.2 m	Low	Coarse Proximity Detection

FINAL DECISION & JUSTIFICATION

Selected Method: Method A. In real-world spatial perception (such as robotic collision avoidance or autonomous driving), error margins directly translate to safety buffers. Method A provides the minimum error tolerance (0.5m), making it the safest choice.

23. Multi-Model Comparison

Model A (88% acc, Fast, Low Rob), Model B (92% acc, Moderate, High Rob), Model C (95% acc, Slow, High Rob). Compare and evaluate the best overall model.

COMPARATIVE EVALUATION

Model A is compromised by low environmental robustness. Model C delivers peak accuracy (95%) and high robustness, but its slow execution speed causes operational latency bottlenecks. Model B combines high accuracy (92%), high environmental robustness, and moderate execution speed.

FINAL RECOMMENDATION

Selected Model: Model B. Model B represents the optimal multi-criteria trade-off, balancing accuracy, speed, and real-world robustness.

24. Motion Detection under Crowd Density

Sparse Crowd: Optical Flow (High), Motion Models (Moderate). Dense Crowd: Optical Flow (Low), Motion Models (High). Compare and evaluate best approach.

COMPARATIVE ANALYSIS

In sparse scenes, optical flow accurately captures fine object contours and pixel vectors without occlusion interference. In dense crowds, severe inter-object occlusion and complex motion boundaries destroy pixel-level brightness constancy assumptions, causing optical flow to fail. Statistical or deep Motion Models perform significantly better in dense scenes by abstracting global crowd movement dynamics.

PROPOSED HYBRID STRATEGY

Selected Strategy: Adaptive Switching / Hybrid Framework. Utilize Optical Flow for low-density crowd monitoring, and automatically switch to structured Motion Models when crowd density exceeds pre-set spatial thresholds.

25. Model Efficiency vs Accuracy

Model A (90% acc, High Comp), Model B (85% acc, Low Comp), Model C (88% acc, Moderate Comp). Compare and evaluate for edge devices.

EDGE DEVICE EVALUATION

Edge devices operate under strict power envelopes, thermal constraints, and limited memory bandwidth. Model A is computationally unviable. Model B maximizes power efficiency but drops accuracy to 85%. Model C delivers a strong 88% accuracy with moderate computational resource demands.

FINAL RECOMMENDATION

Selected Model: Model C (or Model B for ultra-low power devices). Model C is the optimal choice for standard edge platforms (e.g., Jetson Nano, Raspberry Pi 4), capturing 88% accuracy while fitting within device execution budgets.

26. Feature Detection under Illumination Changes

Method A (Normal: 90%, Low Light: 60%), Method B (Normal: 85%, Low Light: 75%). Compare and evaluate robustness.

COMPARATIVE ANALYSIS

Method A experiences a 30% drop under low illumination, indicating reliance on unnormalized intensity gradients (e.g., raw Harris corners or basic intensity templates). Method B maintains high stability across illumination shifts, with only a 10% drop, due to illumination-invariant descriptor design (e.g., SIFT gradient normalization or illumination-robust Deep Features).

FINAL RECOMMENDATION

Selected Method: Method B. Method B is vastly superior for dynamic or unconstrained lighting deployment, maintaining operational reliability regardless of ambient lighting conditions.

27. Video Processing Pipeline Comparison

Pipeline A (88% acc, Fast), Pipeline B (92% acc, Moderate), Pipeline C (95% acc, Slow). Compare and evaluate suitability.

COMPARATIVE ANALYSIS & MAPPING

- **Pipeline A (Fast, 88%):** Tailored for real-time live streaming monitoring, high-speed traffic cameras, or low-latency robotics.
- **Pipeline B (Moderate, 92%):** Ideal for general surveillance systems where minor processing latency (100–200ms) is acceptable for increased accuracy.
- **Pipeline C (Slow, 95%):** Targeted strictly at non-real-time batch processing, post-event video analytics, or forensic evidence collection.

28. Motion Model Evaluation

Model A (Realism: High, Speed: Low), Model B (Realism: Moderate, Speed: High). Compare and evaluate based on application context.

APPLICATION MAPPING

- **Interactive / Real-Time Applications (Gaming, XR Simulators): Select Model B.** High processing throughput is vital to prevent rendering stutter and maintain frame synchronization.
- **High-Fidelity Offline Renderings (Film Production, Scientific Physics Simulations): Select Model A.** Physical visual fidelity takes precedence over computation time.

29. Detection under Occlusion

Method A (70%), Method B (85%), Method C (90%). Compare and evaluate robustness under occlusion.

COMPARATIVE ANALYSIS & EVALUATION

Occlusion disrupts global object features. Method A lacks part-based modeling capabilities, leading to frequent detection drops (70%). Method C retains robust spatial-part representations (e.g., deformable part models or attention-guided deep networks), achieving 90% accuracy under partial visibility.

FINAL DECISION

Selected Method: Method C. Method C provides superior part-based spatial feature learning, ensuring robust target detection in crowded or highly occluded real-world scenes.

30. System Performance under Constraints

Model A (88% acc, 50ms, Low Cost), Model B (92% acc, 80ms, Mod Cost), Model C (95% acc, 120ms, High Cost). Compare and evaluate best system considering all constraints.

Model	Accuracy	Latency	Cost	Overall Score / Suitability
Model A	88%	50 ms	Low	Best Operational Value (Real-Time Budget)
Model B	92%	80 ms	Moderate	Balanced Non-Realtime / Semi-Realtime
Model C	95%	120 ms	High	High-End Offline Solution

FINAL DECISION & JUSTIFICATION

Selected Model: Model A (for budget real-time edge deployment) or Model B (for balanced enterprise surveillance). Model A achieves the ideal balance for real-time production deployment by meeting the strict 50ms real-time latency budget while minimizing deployment costs at an acceptable 88% accuracy. If latency limits permit up to 80ms, Model B delivers a valuable 4% accuracy boost at a moderate cost.